WEEK 6

Forecasting in Pharma: the Patient Funnel

A simple model, well-documented assumptions, and an honest range.

Andi Cupallari · Applied Data Science and AI in Pharma Commercial · Fall 2026 · Monday, October 19

AGENDA

Tonight

0	**What a forecast is for**	Budget vs long-range plan; who uses each
1	**Patient models**	Incidence vs prevalence; flow vs stock
2	**The assumption book**	A forecast is a list of assumptions
3	**From data to assumptions**	Progression, persistence, uptake, share
4	**The forecast**	Funnel → brand → budget and long-range plan
5	**Ranges**	Scenarios, tornado, the plan of record
6	**Checks**	Market trend, regional split, LOE, prevalence model
7	**Close**	Governance · exercise · Homework 2

00

What a forecast is for

Two horizons, one model, many readers.

0 · TWO ASKS

Two questions, one model

LYMBREXA: fictional 3rd-in-class bispecific · 3L+ DLBCL Jan 2027 · 2L label mid-2028

Brand lead — budget
“Monthly patients and revenue for 2027–28. Supply needs vials by month.”

Portfolio strategy — long-range plan
“Annual revenue to 2041. The plan says $250M in 2029, $400M at peak.”



One patient funnel
driven by one assumption book

A launch has no sales history. Its forecast is built from patients and assumptions.

Budget forecast vs long-range plan

	Budget	Long-range plan (LRP)
Horizon	2–3 years	5–15 years, through loss of exclusivity
Grain	monthly; by region and account	annual
Refresh	quarterly	annually, and after major events
Costly error	a miss in the next 12 months	the size and timing of the peak
Owner	brand + finance	portfolio strategy + finance + R&D

The budget is the first two or three years of the LRP, by month. Build one model.

Source: Cook, Forecasting for the Pharmaceutical Industry, 2nd ed. (Gower/Routledge, 2015).

0 · WHO USES IT

Who reads the forecast, and for what

Finance
budget, guidance to investors

Supply
vials by month; plant capacity

Sales operations
targets and incentive pay

Marketing
how much to spend, where

Market access
rebates and contracts

Medical affairs
team size, evidence plans

R&D / business development
which indication next; deal value

Executive team
the plan of record

Most readers are not analysts. Every number must trace to an assumption they can question.

Source: Cook (2015).

01

Patient models

What the funnel starts from, and whether it moves through time.

Incidence and prevalence models start from different patients

I Incidence model

- **Starts from** new cases per year.
- **Fits** oncology lines of therapy, acute care, vaccines.
- **Tonight:** DLBCL.

P Prevalence model

- **Starts from** people living with the disease.
- **Fits** chronic therapy: diabetes, psoriasis, HIV.
- **Tonight:** GLUCOREN in type 2 diabetes.

Linked by duration: prevalence ≈ incidence × how long the disease lasts.

Source: CDC, Principles of Epidemiology, 3rd ed., Lesson 3.

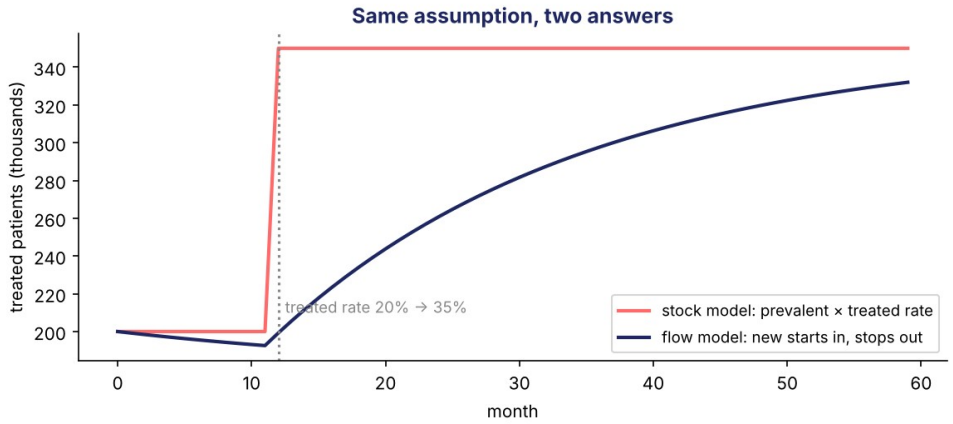
Flow models move patients through time; stock models take a snapshot

Flow model

▸ Patients move from line to line, with a delay. Needed when timing matters: launches, new labels, patent expiry.

Stock model

▸ Patients this year × rates × shares. Each year stands alone. Fine for stable markets.



Same assumption, two answers

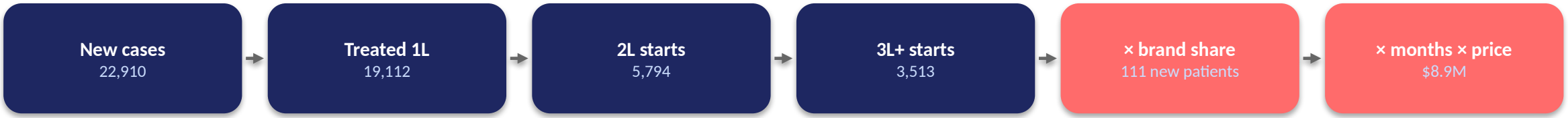
▸ A treated rate rises from 20% to 35%. The stock model jumps; the flow model fills slowly through new starts.

Tonight's DLBCL funnel is a simple flow model: each line feeds the next one a year later.

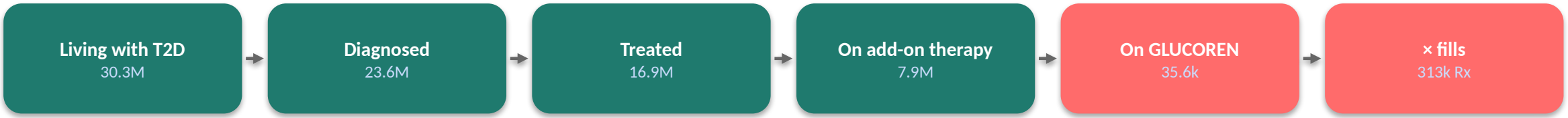
Source: Briggs & Sculpher, PharmacoEconomics (1998). Chart: illustrative.

Three funnels, three different bottlenecks

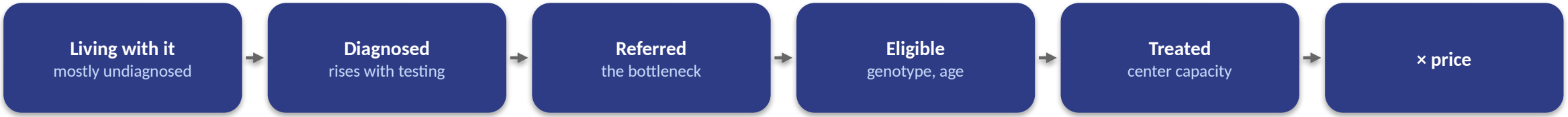
DLBCL — incidence, 2027 (tonight)



Type 2 diabetes — prevalence, 2025 (tonight)



Rare disease — patient finding (for contrast)



Oncology: progression rates. Chronic: treated rate and share. Rare: diagnosis.

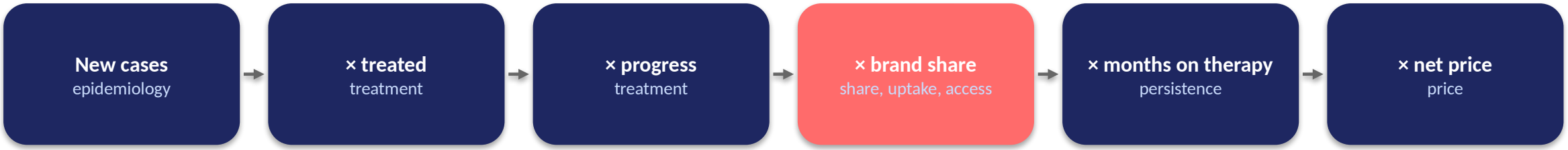
Source: IQVIA, From Orphan to Opportunity (2025). Numbers: synthetic.

02

The assumption book

A forecast is a list of assumptions multiplied together. Write each one down.

A forecast is a chain of assumptions



For every assumption, write down	Example (T2)
Value: low / base / high	26% / 31% / 36%
Source	Week 2 claims cohort, Kaplan–Meier; NEJM review
Reason	trimmed for frail patients who are not re-treated
Owner and date	forecast lead · Oct 2026

The forecast review is a review of the assumption book, not of the output.

Twenty-seven assumptions, each with a range and a source

lymbrexa_assumptions.csv · also the 'Assumptions' sheet of the Excel model

ID	Assumption	Low	Base	High	Source type
E2	Diagnosed and presenting	95%	97%	98%	Data (synthetic)
T1	Treated in 1L	82%	86%	89%	Data (synthetic)
T2	Progress to 2L (lifetime), one year later	26%	31%	36%	Data + literature
T3	Progress to 3L (lifetime), one year later	43%	50%	57%	Data + judgment
T4	Later-lines multiplier (3L+ starts per 3L patient)	1.14	1.24	1.34	Data (synthetic)
B3	Peak share of 3L+ starts	10%	15%	20%	Literature + analog
B4	Peak share of 2L starts	6%	10%	14%	Judgment
U1	Uptake: year 1 (% of peak share)	15%	30%	37%	Analog
U2	Uptake: year 2	60%	90%	95%	Analog
C1	Coverage in launch year	55%	70%	85%	Data + judgment
D1	Average months on therapy	5.0	6.0	7.5	Label + data
P2	Gross-to-net (share of WAC kept)	72%	76%	80%	Judgment
L3	Share retained after the LOE year	25%	40%	55%	Literature + judgment

Source: Full book (27 rows, with sources and reasons) in the CSV and the Excel model.

Five sources of an assumption — say which one you used

Data
claims, market data, analog launches

e.g. progression rates, uptake, persistence

Literature
peer-reviewed studies

e.g. order of entry; LOE erosion; cure rates

Analogs
similar past launches

e.g. how fast share builds

Plans and labels
regulatory dates, label, price

e.g. launch year, months of therapy, price

Judgment
expert view, written down

e.g. 2L peak share, gross-to-net

Judgment is allowed. Undocumented judgment is not.

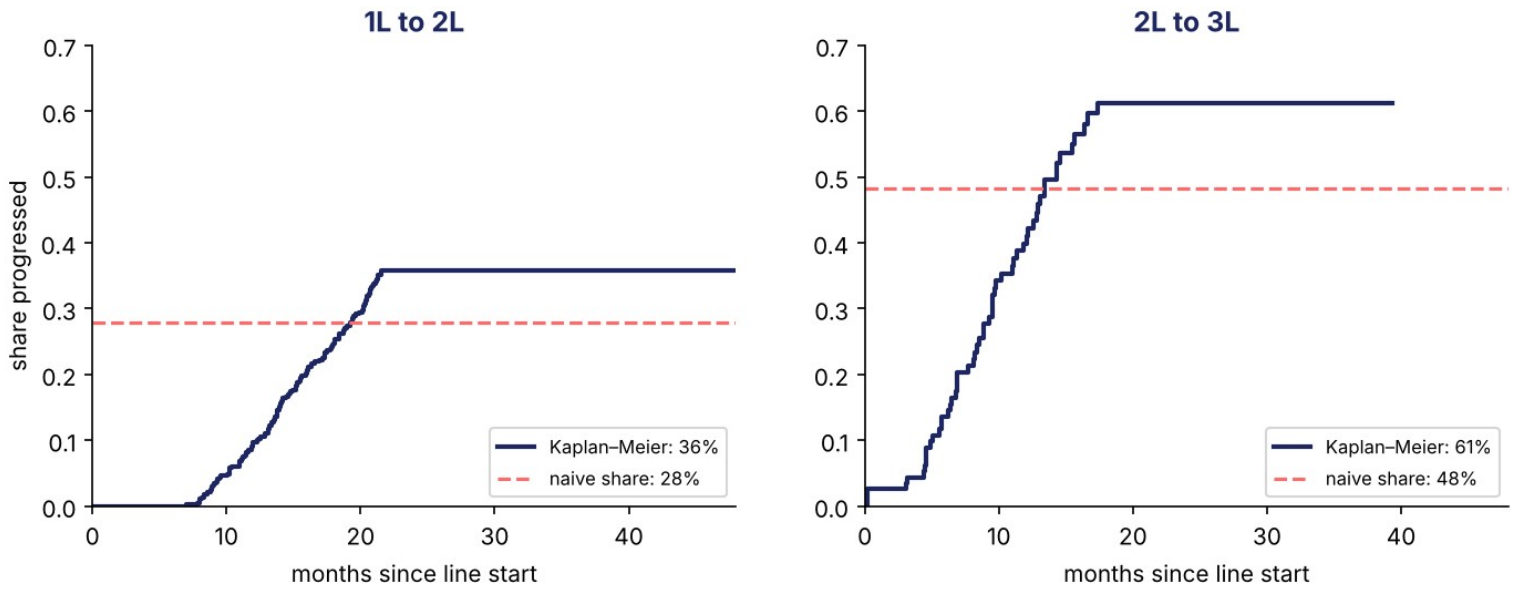
03

From data to assumptions

Use data to justify an assumption, not to fit a complicated model.

Progression rates: count the patients who have not relapsed yet

Week 2 cohort · 482 DLBCL patients · Kaplan–Meier



Naive count

▸ 28% reached 2L. Too low: recent patients have not had time to relapse.

Kaplan–Meier

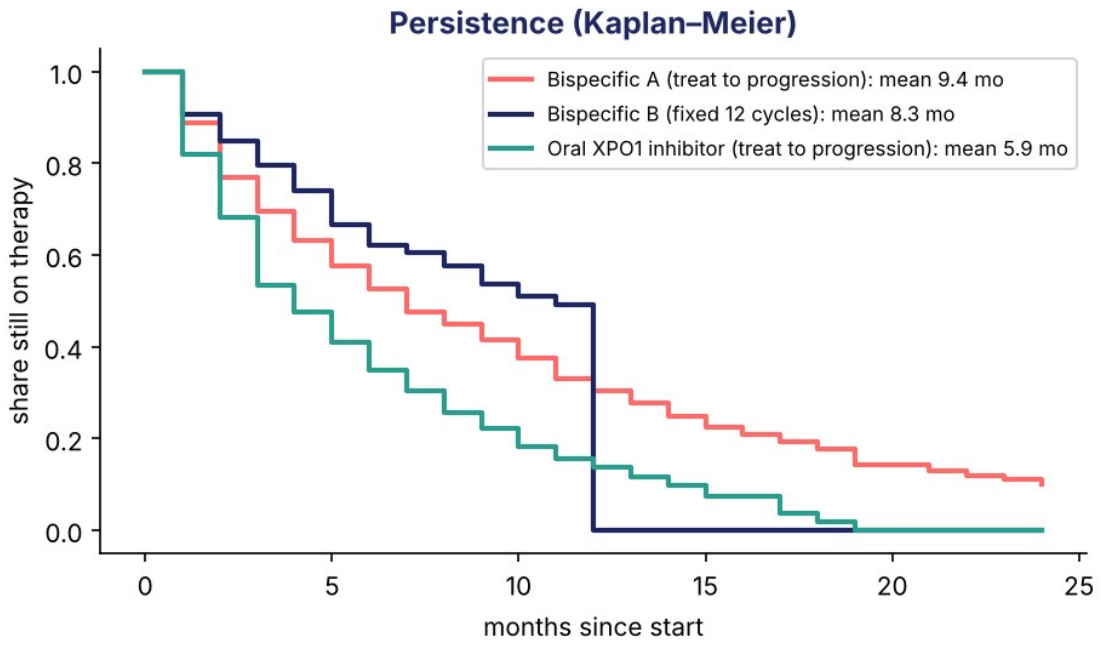
▸ 36% reach 2L over time. Book: 31% (T2), lag one year.

Censored means ‘not yet’, not ‘never’.

Source: Kaplan & Meier, JASA (1958); Sehn & Salles, NEJM (2021): more than 60% cured with first-line therapy. Cohort: synthetic.

Persistence: how many months a patient stays on therapy

Persistence = time from starting to stopping



What the data say

▸ Analog regimens: 5.9 to 9.4 months on average. Naive averages are lower — they ignore patients still on.

Assumption D1

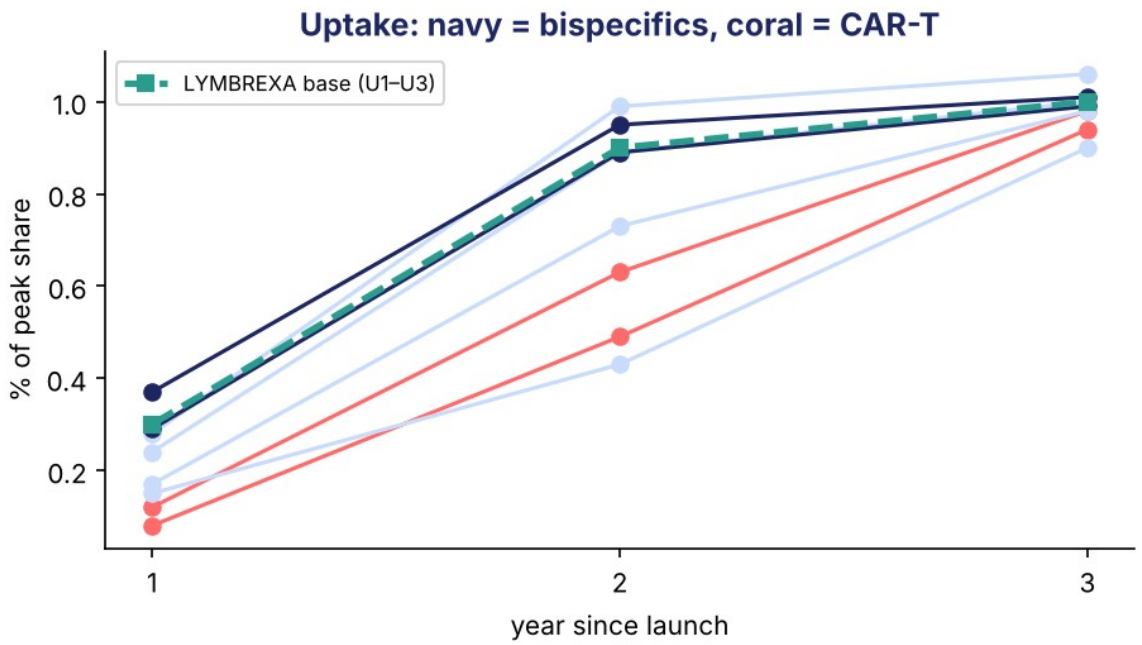
▸ 6 months: the label caps therapy at 8 cycles; some stop early. Range 5–7.5.

Patients on therapy = new patients × months on therapy ÷ 12.

Source: Cramer et al., Value in Health (2008) — definitions; Little, Operations Research (1961). Data: synthetic.

Uptake: read it from analog launches

Share of eventual peak reached in years 1, 2, 3



	Year 1	Year 2	Year 3+
Outpatient analogs	24–37%	89–99%	~100%
CAR-T (cell therapy)	8–12%	49–63%	~95%
LYMBREXA base	30%	90%	100%
LYMBREXA low	15%	60%	95%

Why CAR-T is slower

- 63% of eligible patients are never referred to a treatment center.

No curve-fitting needed: a ramp table from analogs is the assumption.

Source: IQVIA, The Path to CAR T-cell Therapy (2025); Bass, Management Science (1969) for the formal S-curve. Analogs: synthetic.

Peak share: later entrants get less

Entrant share ≈ pioneer share ÷ √(order of entry)

0.26

1st (pioneer)

0.26 ÷ √2 = 0.18

2nd

0.26 ÷ √3 = 0.15

3rd — LYMBREXA

Peak share (B3)

▸ Base 0.15 = the rule. High 0.20 needs evidence the others lack; low 0.10 = a weak follower.

Coverage (C1)

▸ 53% of 3L+ starts come from the top 10% of prescribers. Formulary at the largest centers → 70% reachable in year 1.

Start from the rule; argue away from it only with evidence.

Source: Kalyanaram, Robinson & Urban, Marketing Science (1995) — packaged goods and prescription anti-ulcer drugs. Prescriber data: synthetic.

04

The forecast

Funnel → brand → budget and long-range plan.

The funnel reproduces today's market, then projects it

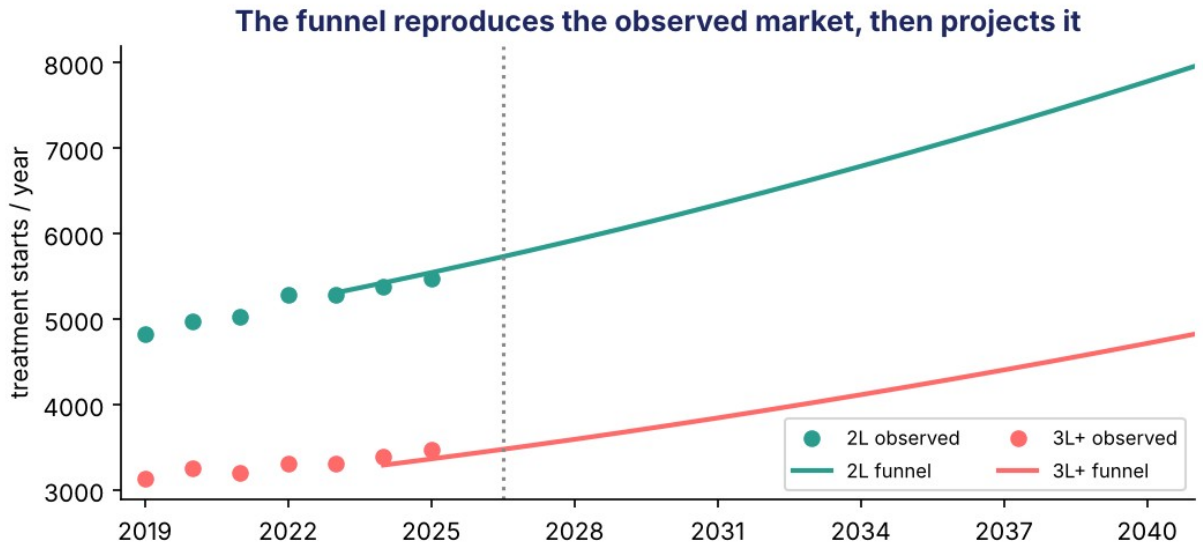
New cases
data, +2%/yr

→ **× diagnosed 97%**

→ **× treated 86%**
= 1L

→ **× 31% a year later**
= 2L

→ **× 50% × 1.24 a year later**
= 3L+



2025	Observed	Funnel
1L	17,930	18,280
2L	5,468	5,544
3L+	3,471	3,363

Within 3% of the observed market. Good enough to forecast on.

Source: dlbcl_market_monthly.csv (synthetic).

From the funnel to LYMBREXA's patients and revenue

2032, base case



New patients vs patients on therapy

- New patients is a flow; patients on therapy is a stock.
- On therapy = new × months ÷ 12 (Little's law).
- The field is paid on new patients; supply plans on patients on therapy.

Net price per month

- $18,500 list × 76% kept after rebates × 95% of doses given = $13,357.

Source: Little, Operations Research (1961). Assumptions: lymbrexa_assumptions.csv.

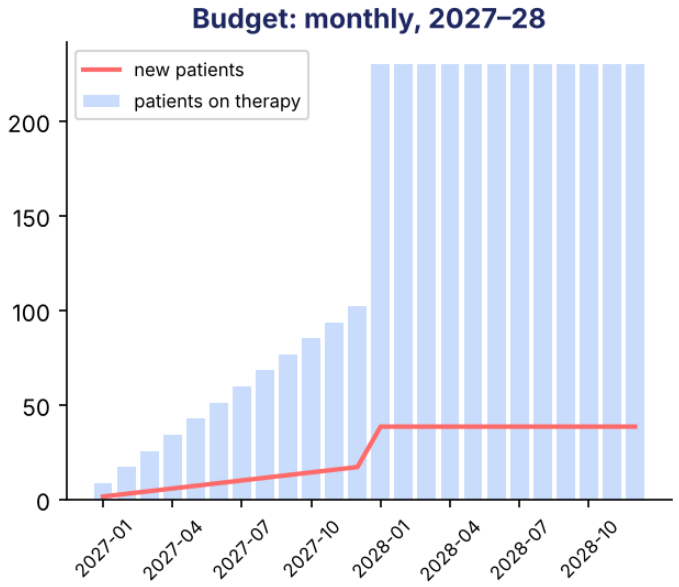
Budget: the first two years, by month

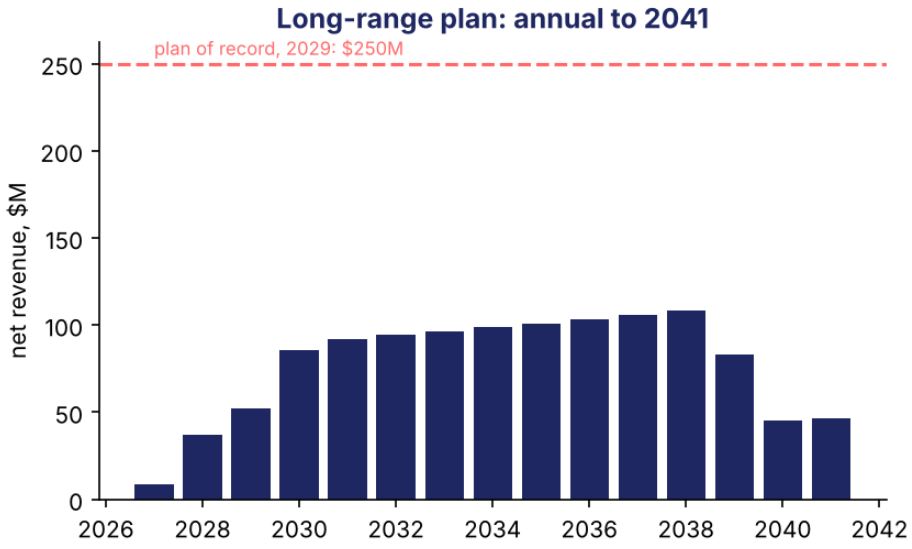
Month	Jan 27	Jun 27	Dec 27	each month of 2028
New patients	1.4	8.5	17.0	38.4
Patients on therapy (≈ new × 6)	8.5	51.1	102.2	230.3
Vials	8.1	48.5	97.1	218.8
Net revenue, $M	0.1	0.7	1.4	3.1

2027: $8.9M. 2028: $36.9M. Launch-year phasing is a straight-line ramp — written down as assumption F1.

Long-range plan: $52M in 2029 and $108M at peak

Plan of record: $250M in 2029, $400M at peak





	2027	2029	2030	2032	2035	2038	2040
New patients	111	651	1,065	1,176	1,259	1,348	564
Net revenue, $M	9	52	85	94	101	108	45

Source: Model output, base case. 2L label adds patients from 2029; patent expiry in 2039.

05

Ranges

Never one number. Show the range and what drives it.

Scenarios: every assumption low, base or high

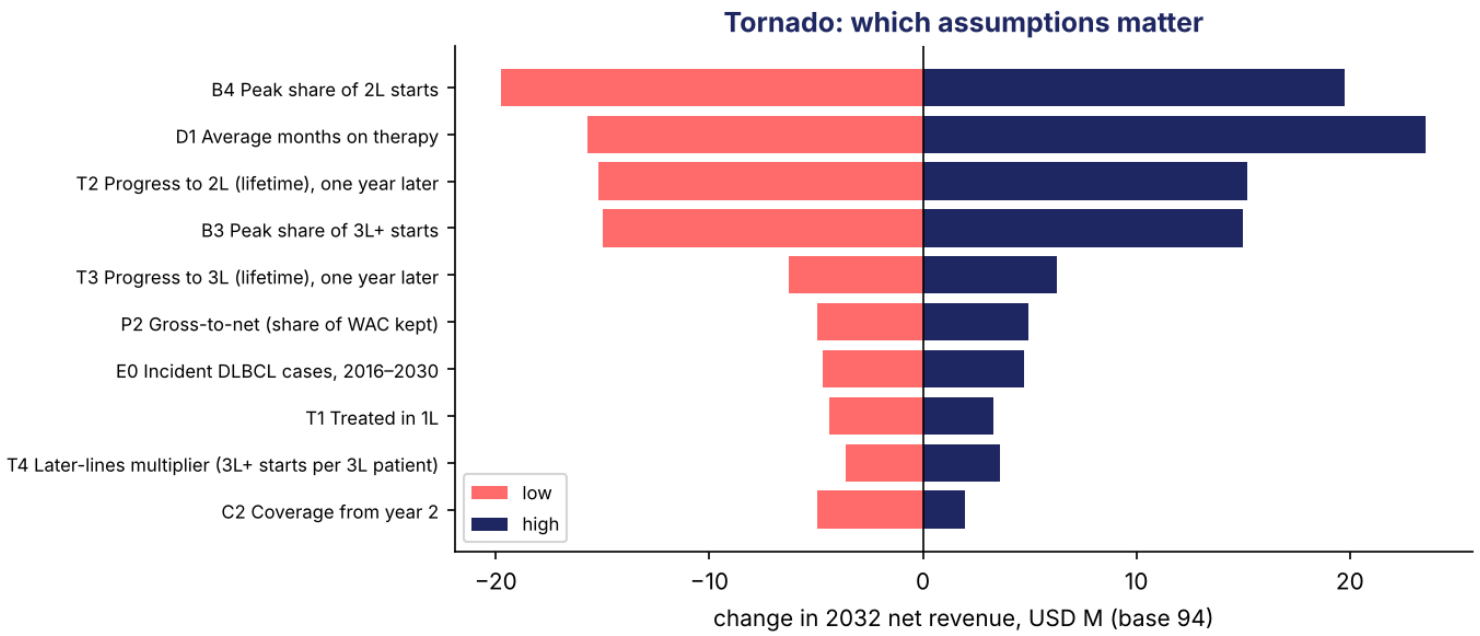
Deliberately extreme — the outer edges, not likely outcomes

Net revenue, $M	2027	2028	2029	2032	2035	Peak
Low	1	7	11	28	30	31
Base	9	37	52	94	101	108
High	37	148	228	251	274	308

Even the all-high case misses the plan's $400M peak.

The tornado shows which assumptions to research

One assumption at a time, low and high · 2032 net revenue, base $94M



Top four

- B4 Peak share of 2L starts: ±$20M
- D1 Average months on therapy: ±$20M
- T2 Progress to 2L: ±$15M
- B3 Peak share of 3L+ starts: ±$15M

Spend the research budget on the top bars: 2L share, months on therapy, progression rates.

What would have to be true for $250M in 2029?

To reach $250M in 2029 you need	Base case	Plausible?
3,119 new patients a year	651	No — 4.8× the base
…which is this share of all 3L+ starts	15% peak	No — 85%
…or 29 months on therapy per patient	6 months	No — label caps at 8 cycles

Honest revision: about $52M in 2029, $108M at peak. $250M needs a different drug.

Source: IQVIA, A Framework for Successful Biopharma Launches (2017): 50–70% of launches miss their first-year forecast.

06

Checks

Simple cross-checks before the forecast leaves your desk.

Check the funnel against a simple trend of the market

3L+ starts · exponential smoothing (ETS) vs ‘same as last year’

Check	Result
Error on the last 12 months (hidden from the model)	ETS 4.6% · same-as-last-year 6.7%
3L+ starts in 2027: trend vs funnel	3,574 vs 3,513 — gap +2%

What time series is for

▸ Projecting a market or an in-line brand for 1–2 years. Not a launch: there is no history.

Foundation models

▸ Chronos, TimesFM: pre-trained, no fitting. Useful for many series at once. Same limit: no history, no forecast.

Source: Hyndman & Athanasopoulos, FPP3 (2021) ch. 8; Ansari et al., TMLR (2024); Das et al., ICML (2024).

6 · SPLIT

Split the national forecast by region and account

Top-down: last year's shares × the national forecast

2028 new patients	Academic centers	Community	Total
Northeast	57	32	90
South	112	63	176
Midwest	59	34	93
West	67	35	102
National			461

Why top-down

- Simple, and every piece adds back to the national number — finance, supply and the field see one forecast.

Source: FPP3 ch. 11 for methods that reconcile forecasts made at every level. Data: synthetic.

Patent expiry: borrow other drugs' history

	Assumption	Basis
Small molecule (GLUCOREN)	brand keeps ~23% of units after one year	Grabowski et al. 2021
Biologic (LYMBREXA)	keeps 75% in the expiry year, 40% after	judgment: biosimilars erode slower

GLUCOREN Rx (thousands)	2026	2027	2028	2029
Trend, no expiry	309	304	300	296
With expiry (Jul 2027)	309	189	69	59

Source: Grabowski, Long, Mortimer & Bilginsoy, J Med Econ (2021). GLUCOREN: synthetic.

GLUCOREN: a prevalence model in one line

Type 2 diabetes · 2025 · each year stands alone



Calibrate

- Brand share of class (4.7%) is set so the model matches 2025 prescriptions.

Then forecast the rates

- Prevalence, treated rate and class share change slowly; write down how much per year.

Source: Rates: synthetic, stylized. CDC, Principles of Epidemiology (prevalence).

07

Close

What the forecaster owns besides the number.

7 · GOVERNANCE

50–70% of launches miss their first-year forecast

Assumption book
every input: range, source, reason, owner

One model
budget = first years of the LRP

Ranges
scenarios and a tornado, every time

Forecast vs actual
track your own misses

Simple checks
funnel vs market; pieces add up

Versions
what changed since last time, and why

The usual miss is too high, not random. A written assumption book is how you catch it.

Source: IQVIA, A Framework for Successful Biopharma Launches (2017), based on ~12,000 launches.

Five common mistakes

Undocumented assumptions
nobody can challenge them

Naive counts
censored patients treated as ‘never’

New patients read as treated
persistence ignored

One number
no range, no tornado

A complicated model
nobody can follow it, so nobody trusts it

7 · EXERCISE

In-class exercise — 20 minutes, groups of 3–4

Week_06_InClass_Exercise.docx

A · Funnel by hand

▸ 2027 inputs → 3L+ starts → new patients → patients on therapy → revenue.

B · Write three assumptions

▸ Peak share, uptake, months on therapy: low/base/high, source, reason.

C · Which model

▸ Six situations: incidence or prevalence; budget or LRP.

Short on time: Part A only.

Homework 2: a forecast for VELTRAXA

Due Monday, November 2, 11:59 PM ET · 10% of grade · data/homework2/

The brand

▸ Fictional oral drug for relapsed mantle cell lymphoma, launch July 2027. Same model, different disease.

Deliverables

▸ The Excel model (or notebook) adapted to VELTRAXA · an assumption book of 20+ rows · a two-page memo.

Assumption book	Funnel + brand	Ranges	Checks	Memo
30	25	15	10	20

7 · NEXT WEEK

Week 7 — NLP in Pharma

Monday, October 26

Before Week 7

- **Read:** Reading Pack B–D; Cook, new-product chapters.
- **Install:** transformers, sentence-transformers.
- **Start:** Homework 2 — the assumption book first.

Bring

- **Your Part B assumptions.**
- **Term project:** your funnel and first ten assumptions.

Office hours Mondays 5:00–6:45 PM · ac5562@columbia.edu

Questions?

1 · A forecast is a chain of assumptions. Write every one down.
2 · Pick the model shape first: incidence or prevalence.
3 · Use data to justify assumptions — handle censoring.
4 · Patients on therapy = new patients × months on therapy.
5 · One model, two horizons, always a range.

Next week — NLP in Pharma · Homework 2 due November 2